

Finding a square root by estimation

It is possible to find a square root through estimation, by choosing a number to multiply by itself, working out the answer, and then altering the number depending on whether the answer needs to be higher or lower.

$$\sqrt{32} = ?$$

$\sqrt{25} = 5$ and $\sqrt{36} = 6$, so the answer must be somewhere between 5 and 6. Start with the midpoint between the two, 5.5:

$$5.5 \times 5.5 = 30.25$$
  Too low

$$5.75 \times 5.75 = 33.0625$$
  Too high

$$5.65 \times 5.65 = 31.9225$$
  Too low

$$5.66 \times 5.66 = 32.0356$$

the square root of 32 is approximately 5.66

this would round down to 32

the square root of 1,000 is approximately 31.62

$$\sqrt{1,000} = ?$$

$\sqrt{1,600} = 40$ and $\sqrt{900} = 30$, so the answer must be between 40 and 30. 1,000 is closer to 900 than 1,600, so start with a number closer to 30, such as 32:

$$32 \times 32 = 1,024$$
  Too high

$$31 \times 31 = 961$$
  Too low

$$31.5 \times 31.5 = 992.25$$
  Too low

$$31.6 \times 31.6 = 998.56$$
  Too low

$$31.65 \times 31.65 = 1,001.72$$
  Too high

$$31.62 \times 31.62 = 999.8244$$

this would round up to 1,000 as the nearest whole number

Finding a cube root by estimation

Cube roots of numbers can also be estimated without a calculator. Use round numbers to start with, then use these answers to get closer to the final answer.

$$\sqrt[3]{32} = ?$$

$3 \times 3 \times 3 = 27$ and $4 \times 4 \times 4 = 64$, so the answer is somewhere between 3 and 4. Start with the midpoint between the two, 3.5:

$$3.5 \times 3.5 \times 3.5 = 42.875$$
  Too high

$$3.3 \times 3.3 \times 3.3 = 35.937$$
  Too high

$$3.1 \times 3.1 \times 3.1 = 29.791$$
  Too low

$$3.2 \times 3.2 \times 3.2 = 32.768$$
  Too high

$$3.18 \times 3.18 \times 3.18 = 32.157432$$

the cube root of 32 is approximately 3.18

this would be 32.2 rounded to the tenths place, which would round to 32

$$\sqrt[3]{800} = ?$$

$9 \times 9 \times 9 = 729$ and $10 \times 10 \times 10 = 1,000$, so the answer is somewhere between 9 and 10. 800 is closer to 729 than 1000, so start with a number closer to 9, such as 9.1:

$$9.1 \times 9.1 \times 9.1 = 753.571$$
  Too low

$$9.3 \times 9.3 \times 9.3 = 804.357$$
  Too high

$$9.27 \times 9.27 \times 9.27 = 796.5979$$
  Too low

$$9.28 \times 9.28 \times 9.28 = 799.1787$$
  Very close

$$9.284 \times 9.284 \times 9.284 = 800.2126$$

the cube root of 800 is approximately 9.284

this would round down to 800